



MULTIMEDIA UNIVERSITY OF KENYA
Riding on Technology, Inspiring Innovation

Multimedia University of Kenya

Operating Systems – II {Semester 1, 2025/2026}

BSc. Computer Science BSc. Software Engineering, BSc. IT and BSc. CT

Task 2: Review Questions – Remote Procedure Calls

1. Justify why RPC systems traditionally favor an **"At-Most-Once"** failure semantic over **"At-Least-Once,"** particularly when executing non-idempotent server procedures, and evaluate the necessary client-side mechanism to guarantee service delivery under network failure.
2. Evaluate the operational suitability of using a **connectionless transport protocol (UDP)** for a high-volume RPC-based distributed database service, and **determine the specific application-layer reliability and ordering mechanisms** that must be implemented to compensate for UDP's lack of guarantees.
3. Appraise **the security model afforded by including an Authentication Verifier** in the RPC message header. **Critique its effectiveness against both replay attacks and man-in-the-middle attacks, and propose a cryptographic enhancement** required for true end-to-end security.
4. Assess the maintenance overhead and performance impact of manually written stubs compared to those generated automatically via an IDL compiler. **Develop a metric to quantify the efficiency** gained by automated stub generation in terms of development time and error rates.
5. Defend the argument that **marshalling/unmarshalling**, despite introducing performance overhead, is a crucial architectural necessity for RPC. Determine a scenario where eliminating this step (e.g., using shared memory for local RPC) would be appropriate and where it would irrevocably break the RPC paradigm.

Release date: November 16, 2025 [17:00] Due by Sunday 23/11/2025 [2359 Hours]